

Tochukwu Elijah Ogri

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Education

University of Florida

Aug. 2024 – Dec. 2026 (Expected)

Ph.D. Mechanical Engineering (GPA: 4.0/4.0)

Gainesville, FL

- Research: Robotics, Nonlinear Control, System Identification, and Learning-based Control

Oklahoma State University

Jan. 2022 – Aug. 2024

Ph.D. Mechanical & Aerospace Engineering

Stillwater, OK

- Doctoral coursework and research in control systems and autonomous systems

University of Benin

Dec. 2014 – Dec. 2019

B.Eng. Mechanical Engineering, First Class Honors

Benin City, Nigeria

Technical Skills

Programming: C++ (Eigen, STL), Python (PyTorch, NumPy, Scikit-learn), MATLAB/Simulink, Linux, Git, L^AT_EX

Robotics & Autonomy: ROS/ROS2, Gazebo, SLAM, Motion Planning, Trajectory Optimization, GNC Systems

Control & Estimation: Adaptive Control, Optimal Control (LQR/MPC), Control Barrier Functions (CBF), Lyapunov Stability, Kalman Filtering (EKF/UKF), System Identification (SINDy, ICL), Gaussian Processes

Design Tools: SolidWorks

Experience

Graduate Research Assistant

Aug. 2024 – Present

Systems, Cognition, and Control (SCC) Lab, University of Florida

Gainesville, FL

- Developed real-time control architectures combining model-based Reinforcement Learning and Adaptive Optimal Control for nonlinear robotic systems; deployed on UAV/UGV platforms (Parrot Bebop, TurtleBot, Husarion) using C++ and ROS.
- Designed safety-critical controllers using Control Barrier Functions (CBFs) and Quadratic Programming (QP) to enforce state constraints and collision avoidance in real-time autonomous systems.
- Implemented sparsity-promoting system identification (Lasso-based) to learn unknown robotic dynamics from streaming data.

Robotics & Autonomy Intern

May 2024 – Jul. 2024

Autonomous Vehicles Lab, University of Florida Research and Engineering Education Facility

Shalimar, FL

- Designed Guidance, Navigation, and Control (GNC) architectures and trajectory optimization frameworks for the U.S. Air Force Research Laboratory (AFRL) *Safe Trusted Autonomy for Responsible Spacecraft (STARS)* program for orbital inspection under illumination, occlusion, and visibility constraints.
- Implemented sequential feature tracking and real-time motion planning algorithms, validated in hardware-in-the-loop (HIL) simulation for robustness under sensor-denied and deployment-relevant conditions.

Robotics & Autonomy Intern

May 2023 – Aug. 2023

Autonomous Vehicles Lab, University of Florida Research and Engineering Education Facility

Shalimar, FL

- Designed monocular vision observers and extended Kalman filter (EKF) modules, integrating them into ROS2 software stacks to enable SLAM in GPS-denied environments.
- Executed AFRL-sponsored flight experiments across UAV (Parrot Bebop) and UGV (TurtleBot) platforms to validate algorithmic performance under varying environmental conditions.

Graduate Research Assistant

Jan. 2022 – Aug. 2024

Systems, Cognition, and Control (SCC) Lab, Oklahoma State University

Stillwater, OK

- Developed output-feedback adaptive controllers with formal stability guarantees, reducing sensing requirements in trajectory tracking problems.
- Applied Gaussian Process-based error correction for robotic field mapping, improving localization accuracy by fusing EKF estimates with spatial priors.

Software Engineering - Graduate Trainee

May 2021 – Dec. 2021

Ecobank Transnational Incorporated

Lagos, Nigeria

- Built backend services in C#/.NET and Java (Spring Boot) for high-throughput financial systems, focusing on low-latency processing and reliability.

Selected Publications

- T. E. Ogri, M. Qureshi, Z. I. Bell, and R. Kamalapurkar, *State and input constrained output-feedback adaptive optimal control of affine nonlinear systems*, *Automatica*, 2026
- M. Qureshi, T. E. Ogri, H. Ramos, W. A. Makumi, and R. Kamalapurkar, **A Switched Adaptive Control Approach to Reduce Sensing Needs in Trajectory Tracking Problems**, *IEEE Control Systems Letters*, 2025.
- T. E. Ogri, M. Qureshi, Z. I. Bell, W. A. Makumi, and R. Kamalapurkar, **An Output Feedback Approach to Differential Graphical Games in Linear Multiagent Systems**, *IEEE Control Systems Letters*, 2025.
- T. E. Ogri, M. Qureshi, Z. I. Bell, M. Longmire, and R. Kamalapurkar, **Monocular Inspection of Spacecraft Under Illumination Constraints and Avoidance Regions**, *IEEE American Control Conference (ACC)*, 2025.
- T.E. Ogri, M. Qureshi, Z.I. Bell, K. Waters, and R. Kamalapurkar, **An adaptive optimal control approach to monocular depth observability maximization**, *IEEE ACC*, 2024.